LINEAR REGRESSION

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Linear Regression

The following concepts will be introduced:

- ✓ **Simple linear regression**
- ✓ **Multiple linear regression**
- ✓ **Residuals**
- ✓ **Model checking**
- ✓ **Multi-collinearity**
- ✓ **Confidence e prediction bounds**
- ✓ **Variable selection**

They aim to identify a functional relationship between a target attribute and a set of explanatory attributes. They allow for the discovery of the existing relationship between the explanatory variables and the target variable, and at the same time, they enable predictions about the values assumed by the target variable for given fixed values of the explanatory variables

Let's suppose we have a dataset containing 'n+1' attributes, 'n' attributes that are candidates to be explanatory, and one target attribute, and 'm' observations.

The target is often referred to as the response variable, output, or dependent variable, while the explanatory attributes are usually mentioned as independent variables or predictors.

The realization of the independent variables for each observation 'i' is denoted by $\boldsymbol{x_i}$, while the realization of the target attribute is denoted by y_i.

Additionally, we denote by **X** the matrix, of dimensions [m×n], of the realizations of the 'n' independent variables and by **y**=(y_1, ..., y_m). Finally, we denote by ***Y*** the random variable associated with the target variable and by ***X^j*** the j-th explanatory variable.

In these models we assume that there exists a function

$$f : R^n \rightarrow \mathrm{R}$$

Describing the relationship between the dependent variable ***Y*** and the "n" independent variables ***X^j***, we assume

$$Y = f\left(X^1, \ldots, X^n\right)$$

In reality we assume the following

$$Y = f\left(X^1, \ldots, X^n\right) + \varepsilon$$

SIMPLE LINEAR REGRESSION

Linear regression models represent the most well-known class of estimation models and are based on a hypothesis space consisting only of linear functions. Consequently, the functional relationship mentioned earlier reduces to:

$$f\left(X^1,\ldots,X^n\right) = w^1X^1 + \ldots + w^nX^n + b = \sum_{j=1}^{n} w^jX^j + b$$

In the case where there is only one independent variable "***X***" (n=1), the model is called a simple linear regression model and reduces to the following relationship

$$f(X) = wX + b$$

This model is particularly important, not so much for its actual use in real-world applications, but because it allows us to present the main characteristics of linear regression models with a generic number of candidate explanatory variables, which we will introduce and use later.

Simple linear regression

Let's consider a database that contains information about the sales volumes of large computer outlets.

Sales:	sells volume
Titles:	number of different product types
Footage:	outlet dimension
IBMbase:	number of IBM customers
Appbase:	number of APPLE customers

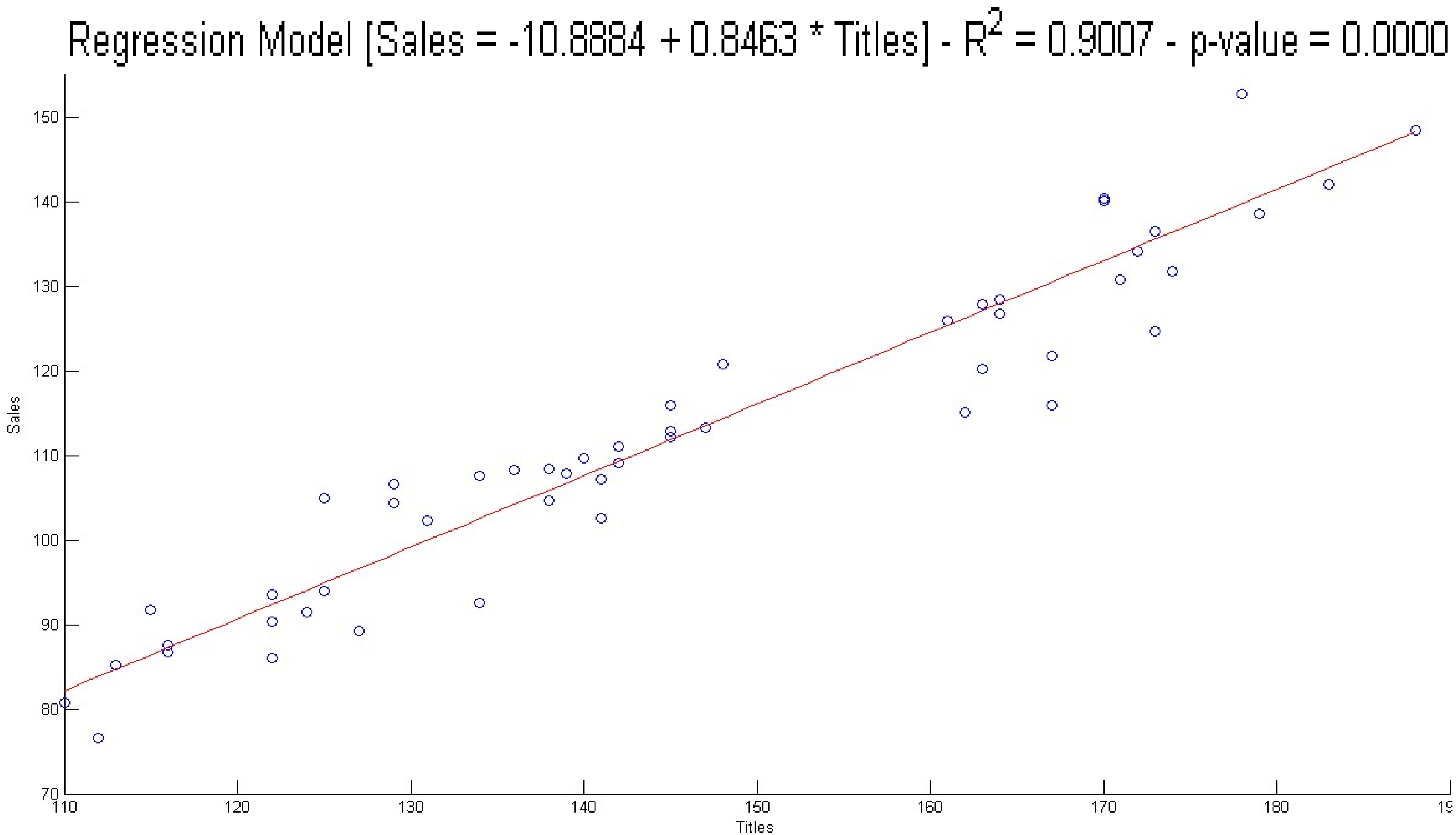
Let's suppose we want to model the sales volume value (Sales) as a function of the number of different software items on display (Titles). We will use the following simple linear regression model.

$$Y = wX + b + \varepsilon \longrightarrow \text{Sales} = w\,\text{Titles} + b + \varepsilon$$

where "ε", know as residual, is defined as follows

$$\varepsilon = Y - f(X)$$

SIMPLE LINEAR REGRESSION



The relationship being sought is approximate in nature, and therefore it is essential to establish a criterion by which to identify, among all the lines in the plane, the line that best expresses the relationship between ***Y*** and ***X***.

The most well-known and widely used criterion for determining the regression coefficients "***w***" and "***b***" consists of minimizing the sum of squared errors, expressed by the following Sum of Squared Errors (SSE) function:

$$SSE = \sum_{i=1}^{m} e_i^2 = \sum_{i=1}^{m} [y_i - f(x_i)]^2 = \sum_{i=1}^{m} [y_i - wx_i - b]^2$$

This function is quadratic and convex with respect to the parameters of the regression model and therefore admits a unique minimum point, which can be obtained by setting its partial derivatives with respect to the two parameters of the regression model to zero.

In this specific case, it is sufficient to solve a linear system of 2 equations in 2 unknowns to obtain the optimal values of the regression model parameters

Let's turn our attention to the general case where "n>1" explanatory variables are considered simultaneously. In this case, the probabilistic relationship between the target variable "***Y***" and the 'n' explanatory variables "X^j" will be:

$$Y = w^1 X^1 + \ldots + w^n X^n + b + \varepsilon$$

Returning to the previously presented dataset, we use the following variables.

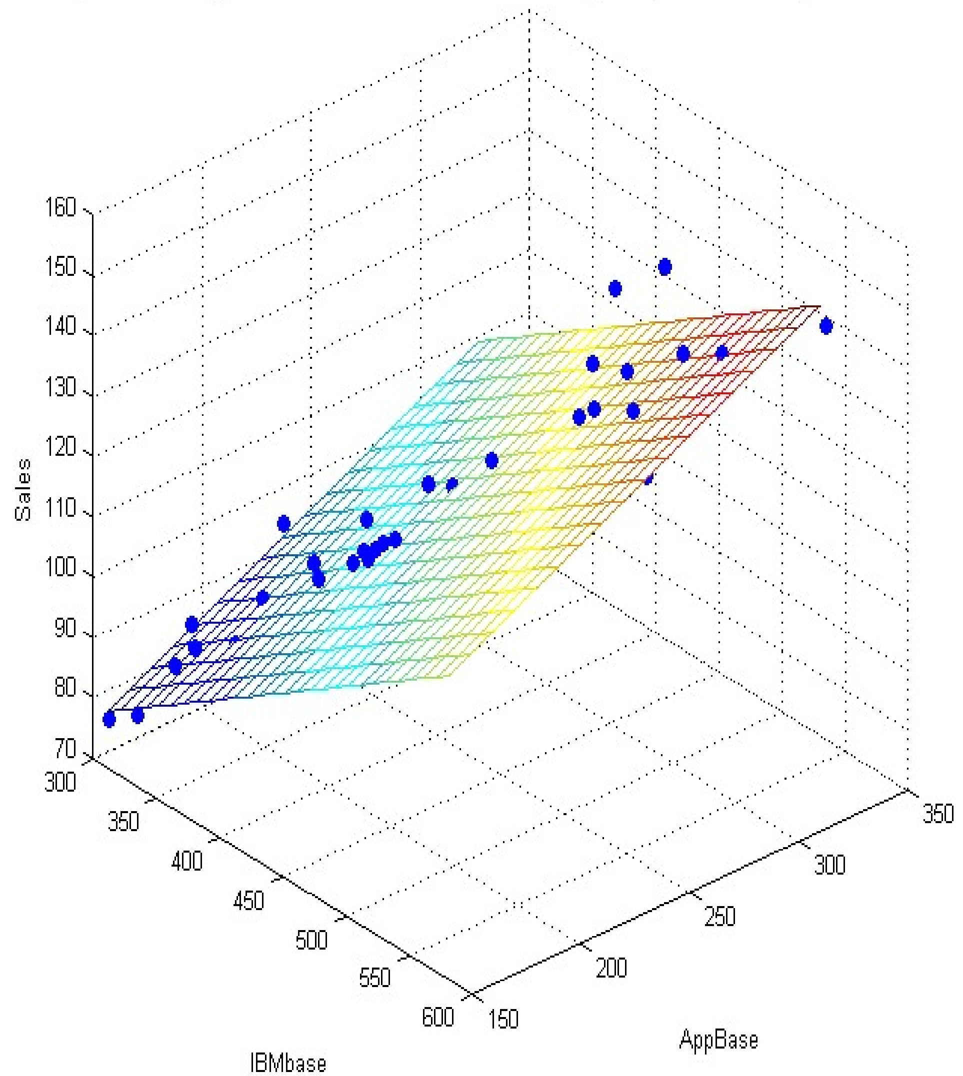
- **IBMbase**
- **Appbase**

To predict the value of **Sales**, by using the following model

$$\text{Sales} = w^{\text{IBMbase}} \text{IBMbase} + w^{\text{AppBase}} \text{Appbase} + b + \varepsilon$$

Multiple linear regression

Regression Model [Sales = 2.19 + 0.15 * IBMbase + 0.19 * AppBase] - R^2 = 0.8834 - p-value = 0.0000



It is important to emphasize how the multiple linear regression model can be generalized to take into account not only the appearance of individual terms but also their power elevation and the rectangular or interaction terms of higher order.

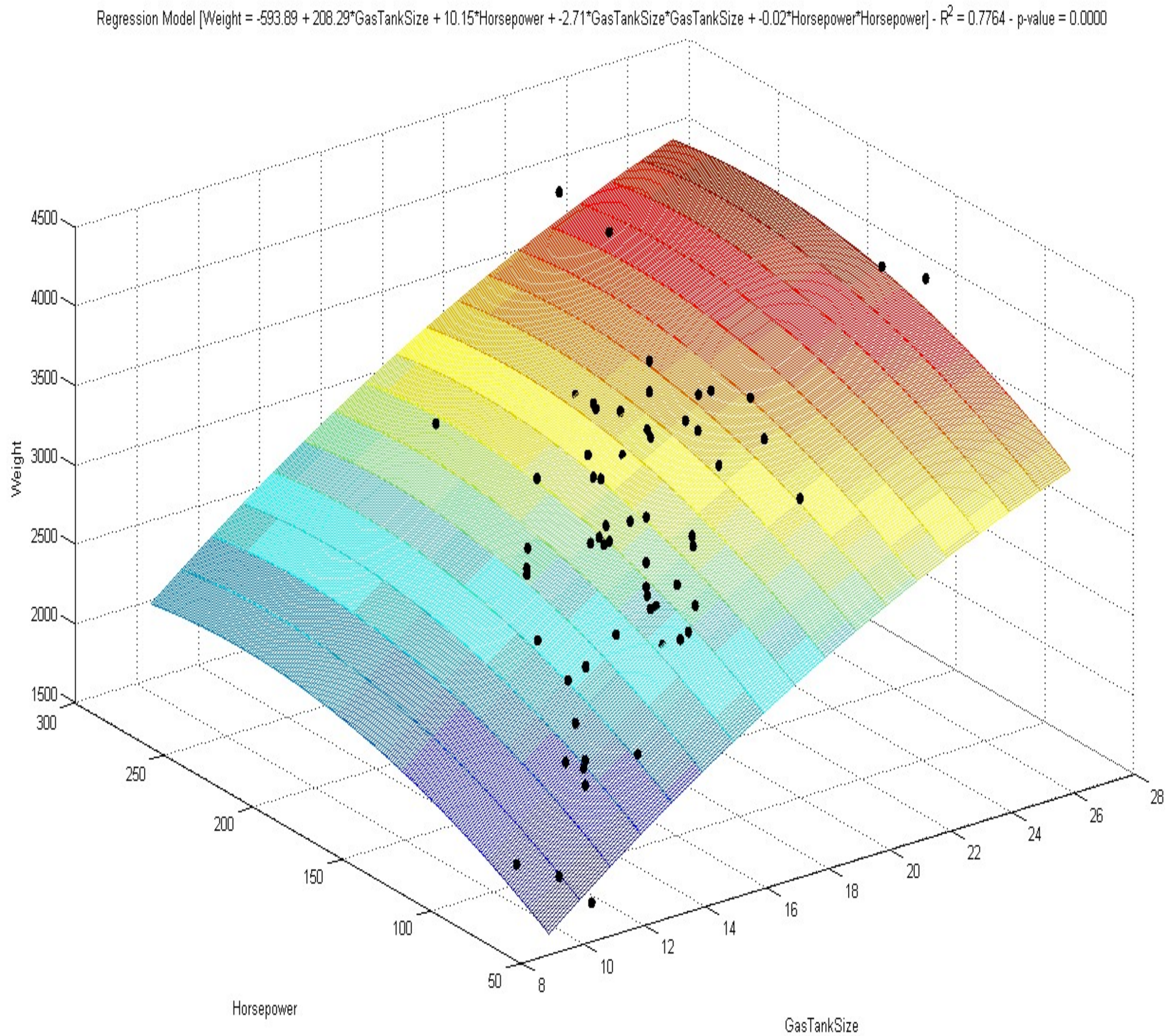
In this case the probabilistic relationship between the target variable "Y" and the "n" explanatory variables "X^j" is described as follows:

$$Y = \boxed{w^1 X^1 + \ldots + w^n X^n} + \boxed{w^{n+1} \left(X^1\right)^2} + \boxed{w^{n+2} X^1 X^2} + \ldots + w^l \left(X^n\right)^2 + \boxed{b + \varepsilon}$$

The number of parameters to be estimated by minimizing the SSE function, is given by "l+1" which is also known as the degrees of freedom of the multiple linear regression model.

MULTIPLE LINEAR REGRESSION

Regression Model [Weight = -593.89 + 208.29*GasTankSize + 10.15*Horsepower + -2.71*GasTankSize*GasTankSize + -0.02*Horsepower*Horsepower] - R^2 = 0.7764 - p-value = 0.0000



Assumptions about residuals

The random variable "ε" representing the errors, appearing in the regression models we have previously discussed, must satisfy certain assumptions.

The variable "ε" must be comparable to random noise, without observable effects on the response variable "Y". Usually, "ε" is thought of as the result of the effect of different independent variables, excluded from the regression model, each of which contributes a negligible effect on the values assumed by the target attribute "Y".

If the determination of the regression coefficients is done by minimizing the SSE function, the random variable "ε" must be distributed according to a normal distribution with zero mean and standard deviation "σ". This assumption translates into the following first hypothesis

$$\boldsymbol{E}(\boldsymbol{\varepsilon}_i \mid \boldsymbol{x}_i) = 0 \qquad \boldsymbol{Var}(\boldsymbol{\varepsilon}_i \mid \boldsymbol{x}_i) = \sigma^2$$

ASSUMPTIONS ON RESIDUALS

The second hypothesis requires that the residuals "ε_i" and "ε_k", corresponding to two distinct observations "$\boldsymbol{x_i}$" and "$\boldsymbol{x_k}$", are independent for any choice of "i" and "k".

Finally, the third hypothesis requires that the standard deviation (variance) of the residuals remains constant, regardless of the values of the individual components of the observations. This is referred to as homoscedasticity (heteroscedasticity in the opposite case

We also note that the regression model is more accurate the closer the standard deviation of the residuals (standardized) is to zero. Typically, the estimator of the standard deviation, known as the standard error of the estimate, is computed and plays a very important role in determining the accuracy of a regression model, as its value determines the extent of uncertainty (dispersion) relative to the predicted value.

MODEL ASSESSMENT

The setup and development of a linear regression model may seem simple, since even the most popular spreadsheets allow for the determination of the coefficients. However, the ease of processing can foster false certainties about the actual significance of the derived estimation model

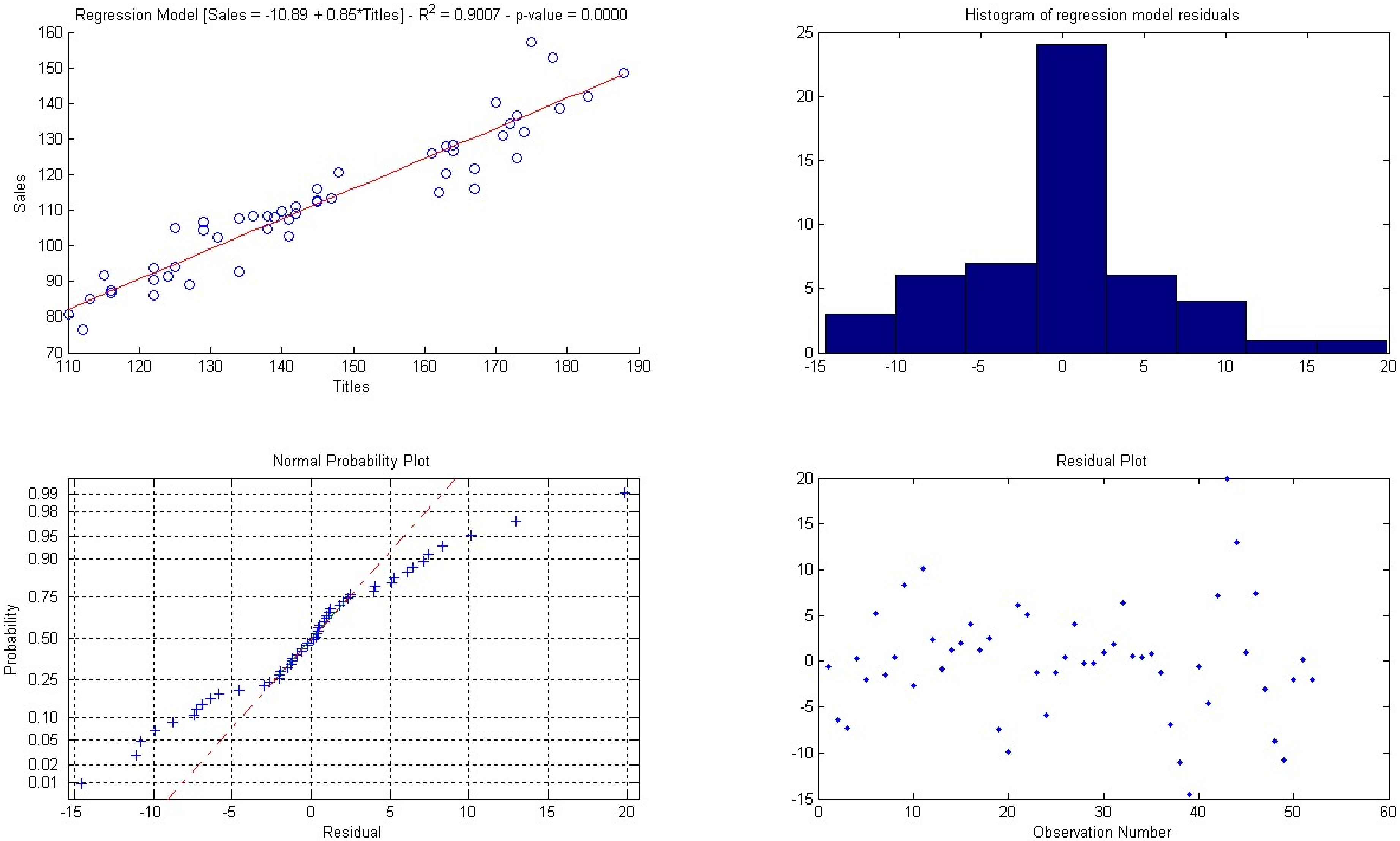
There are several aspects that we will consider to evaluate the quality and predictive accuracy of a linear regression model:

- *Normality and independence of residuals*
- *Significance of coefficients*
- *Analysis of variance*
- *Coefficient of determination*
- *Linear correlation coefficient*
- *Multicollinearity of explanatory variables*
- *Confidence limits of the prediction*

MODEL ASSESSMENT

Normality and independence of residuals

To verify if the distribution of the residuals is normal, we have graphical representations and hypothesis tests available.



Model assessment

Appropriate hypothesis test are:

- *Kolmogorov-Smirnoff*
- *Shapiro-Wilk*
- *Lilliefors*

The results of the hypothesis tests must be interpreted in light of what emerges from the analysis of the graphs

Kolmogorov-Smirnoff normality test on residuals REJECTED at the 0.05 level

Lilliefors normality test on residuals REJECTED at the 0.05 level

Kolmogorov-Smirnoff normality test on residuals REJECTED at the 0.01 level

Lilliefors normality test on residuals REJECTED at the 0.01 level

Model assessment

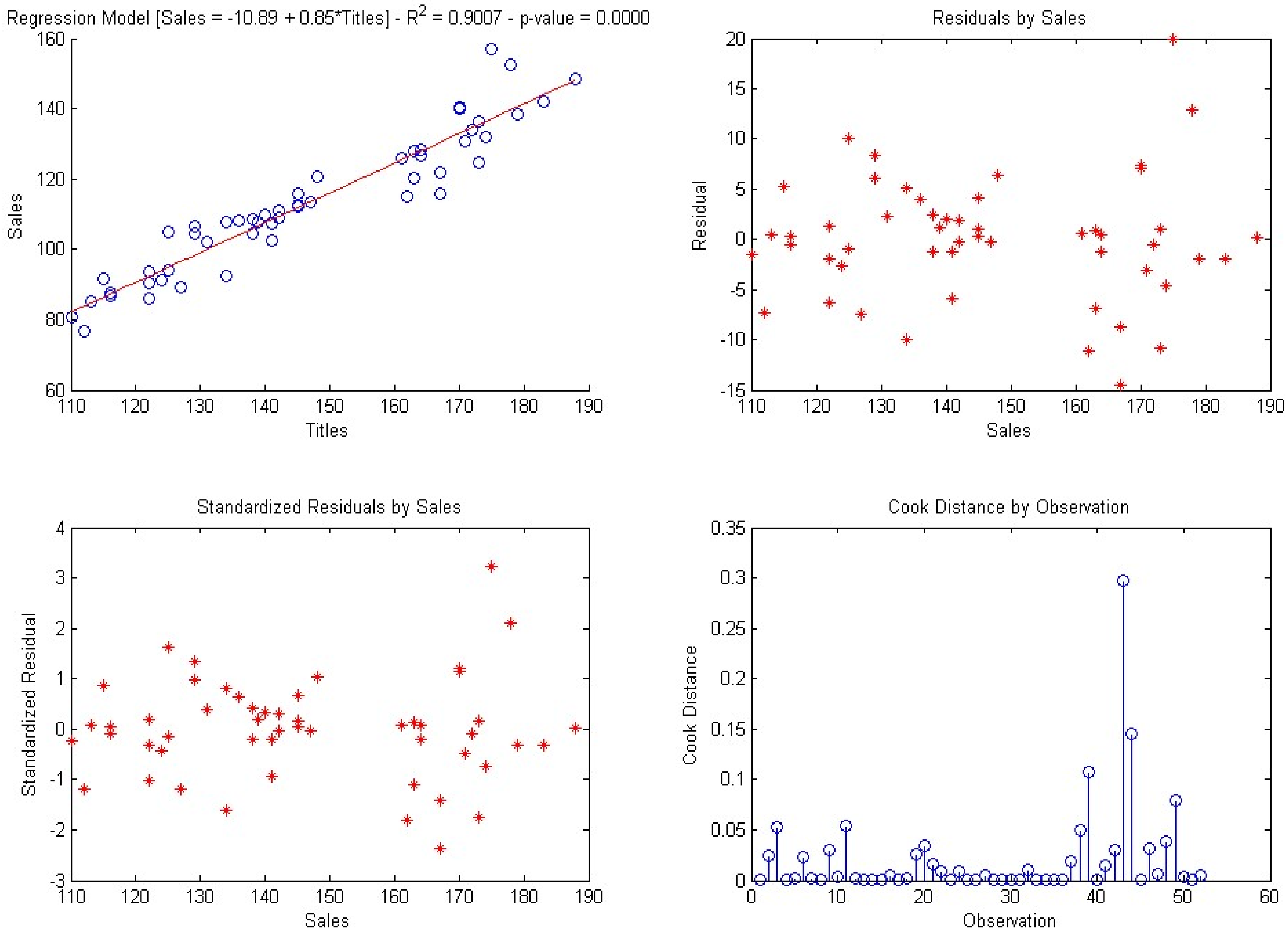
To verify if the residuals "ε_i" and "ε_k" are independent for any choice of "i" and "k", we have an appropriate hypothesis test available.

This is the Durbin-Watson test which, when applied to the residuals of the regression model related to **Sales**, leads to rejecting the hypothesis of independence of the residuals, with a p-value of 0.0039

Homoscedasticity of the residuals can be evaluated graphically.

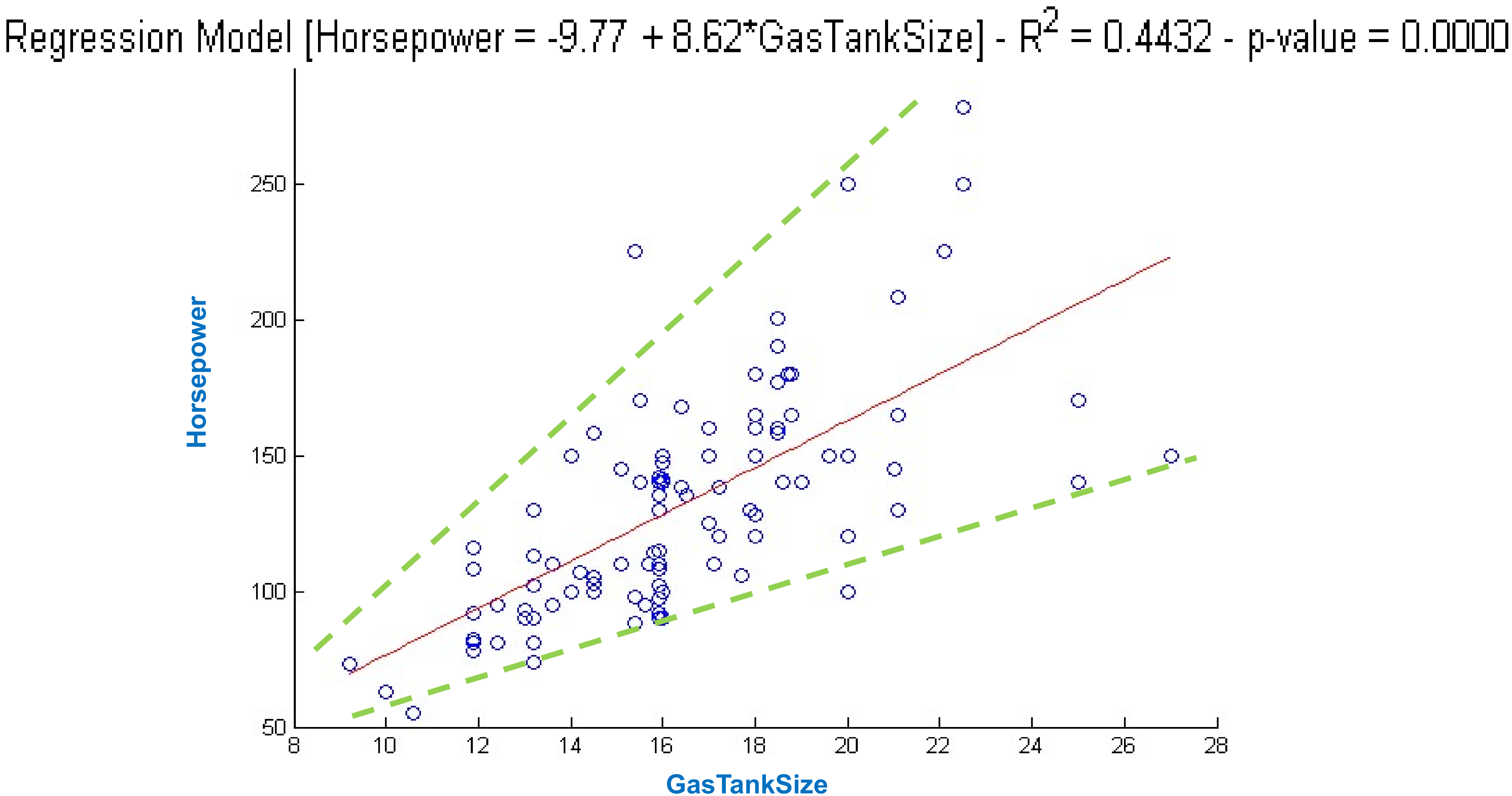
Additionally, it is possible to assess the influence of each observation on the developed regression model; this influence is measured by Cook's distance, which is usually denoted as "D_i" for each residual "i". Generally, values of Cook's distance greater than one lead to considering the corresponding observation as influential and thus to be analyzed with particular attention for the purpose of validating the considered linear regression model.

MODEL ASSESSMENT

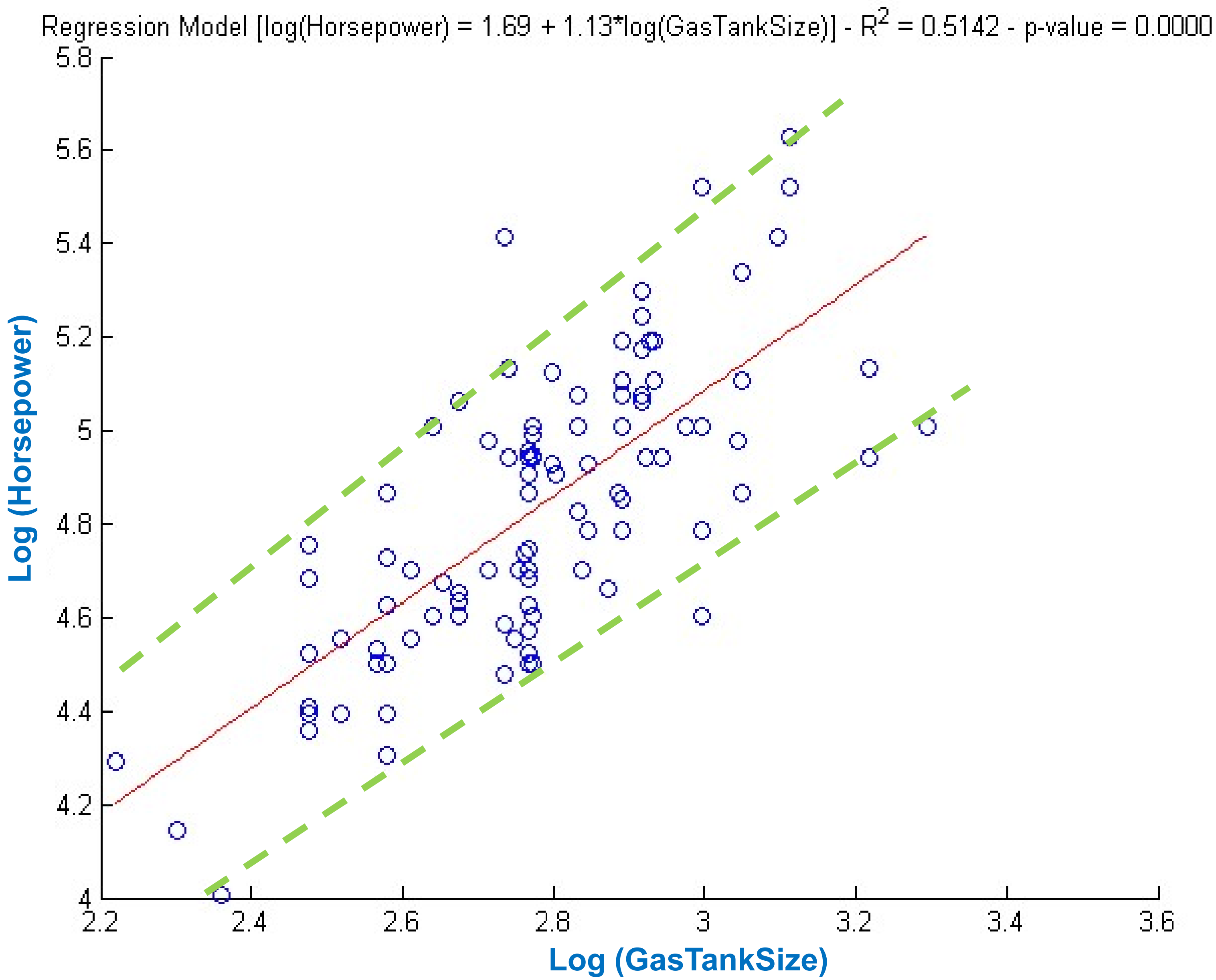


Model assessment

Homoscedasticity of the residuals can be evaluated graphically



The lack of homoscedasticity compromises the value of the regression model and must be absolutely corrected, usually by applying appropriate data transformations (logarithmic transformation applied to both axes)



Coefficient Significance

The values

$$\hat{w}$$

obtained by minimizing SSE, are not the true values of the regression coefficients

$$w = \left(w^1, \ldots, w^{l+1}\right)$$

appearing the regression model,

$$Y = f\left(X^1, \ldots, X^n\right) = w^1 X^1 + \ldots + w^n X^n + w^{n+1}\left(X^1\right)^2 + w^{n+2} X^1 X^2 + \ldots + w^l \left(X^n\right)^2 + b + \varepsilon$$

but they are an estimate for them.

We are faced with a statistical inference problem where regression coefficients are the unknown parameters of the population, the "m" observations represent a sample from the population and the coefficient minimizing SSE are point estimators which are typically associated with a confidence interval.

MODEL ASSESSMENT

It is possible to evaluate the importance of each explanatory variable in predicting the value of the response variable by assessing the confidence interval of the corresponding regression coefficient.

If the confidence interval contains the null value, there is no reason to believe that the corresponding explanatory variable is indeed significant for the response variable in the linear regression model.

```
-------------------------------------------------------
Coefficients C.I. at 0.90 confidence level
-------------------------------------------------------
        intercept =   4.154318 [ -0.170000,   8.478636]
           Weight =   0.003694 [  0.002281,   0.005108]
       Horsepower =  -0.008614 [ -0.047757,   0.030529]
Weight*Horsepower =   0.000006 [ -0.000006,   0.000017]
-------------------------------------------------------


-------------------------------------------------------
Coefficients C.I. at 0.95 confidence level
-------------------------------------------------------
        intercept =   4.154318 [ -1.011619,   9.320256]
           Weight =   0.003694 [  0.002005,   0.005383]
       Horsepower =  -0.008614 [ -0.055375,   0.038147]
Weight*Horsepower =   0.000006 [ -0.000008,   0.000019]
-------------------------------------------------------
```

MODEL ASSESSMENT

It is possible to evaluate the importance of each explanatory variable in predicting the value of the response variable by assessing the confidence interval of the corresponding regression coefficient.

If the confidence interval contains the null value, there is no reason to believe that the corresponding explanatory variable is indeed significant for the response variable in the linear regression model.

probability to observe a t-value greater than this by pure chance

statistic associated to hypothesis test

significativity

```
-----------------------------------------------------------------
Regression Model Summary
GasTankSize = a0 + a1*Weight + a2*Horsepower + a3*Weight*Horsepower
-----------------------------------------------------------------
Termine      Stima       Errore Std     t-value    Pr > |t|
   a0      4.154318     2.607253        1.5934     0.1139
   a1      0.003694     0.000852        4.3343     0.0000
   a2     -0.008614     0.023600       -0.3650     0.7158
   a3      0.000006     0.000007        0.8268     0.4101
-----------------------------------------------------------------
```

Analysis of Variance

Analysis of variance (**ANOVA**, **AN**alysis **O**f **VA**riance) represents a tool for evaluating the relationship linking the response variable to the categorical explanatory variables.

The information characterizing ANOVA are as follows:

Sum Sq.	sum of squares	$RSS_{reg} = \sum_{i=1}^{m} (\hat{y}_i - \bar{y})^2$ $\quad RSS_{tot} = \sum_{i=1}^{m} (y_i - \bar{y})^2$
d.f.	degrees of freedom of the specific component	
Mean Sq.	$\frac{RSS}{d.f.}$	
F	values of the statistic $F = \dfrac{\frac{RSS}{d.f.}}{\frac{\sum_{i=1}^{m} (\hat{y}_i - y_i)^2}{d.f.err}}$	
Prob>*F*	probability to observe a value greater than *F by pure chance*	

Model assessment

The analysis of variance for the regression model just analyzed leads to obtaining:

Regression ANOVA

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
Model	791.67	3.00	263.89	99.81	0.00
Resid	296.13	112.00	2.64		
Total	1087.79	115.00			

Model assessment

In the case where the ANOVA of the response variable Gas Tank Size is considered with respect to the explanatory variables Type and Country, we will obtain the following

Japan, Other, USA

Small, Medium, Compact, Large, Sporty

Analysis of Variance

Source	Sum Sq.	d.f.	Mean Sq.	F	Prob>F
Country	0.87	2	0.4374	0.11	0.9
Type	331.51	4	82.8779	19.98	0
Country*Type	100.62	8	12.5779	3.03	0.0043
Error	418.93	101	4.1478		
Total	1087.79	115			

Constrained (Type III) sums of squares.

In the case under consideration, it is possible to conclude that the variable Type leads to different mean values of the target variable Gas Tank Size, while the variable Country does not seem to lead to different mean values of the target variable

Coefficient of determination

The coefficient of determination, also known as multiple R^2, represents the percentage of total variance explained by the predictive variables, and thus by the model

$$R^2 = \frac{RSS_{reg}}{RSS_{tot}} = \frac{\sum_{i=1}^{m}\left(\hat{y}_i - \overline{y}_i\right)^2}{\sum_{i=1}^{m}\left(y_i - \overline{y}_i\right)^2} = 1 - \frac{\sum_{i=1}^{m}\left(\hat{y}_i - y_i\right)^2}{\sum_{i=1}^{m}\left(y_i - \overline{y}_i\right)^2} = 1 - \frac{RSS_{err}}{RSS_{tot}}$$

It takes values in the range [0,1], and if for a model the value is very close to one, then the regression model explains a large part of the variation of the target variable (Y). The coefficient of determination tends to increase with the number of variables included in the linear regression model (number of parameters associated with the regression model, i.e., degrees of freedom “l+1”) and tends to overestimate the percentage of variance explained by the model.

To avoid this inconvenience, another indicator is normally used, known as adjusted R^2, defined as follows

$$R^2_{adjusted} = 1 - \left(1 - R^2\right)\frac{m-1}{m-l-1}$$

In the case of the regression model for the target variable presented earlier, we will have

$$R^2 = 0.72777$$

$$R^2_{adjusted} = 0.72048$$

In the case under consideration, there is no significant difference between the two coefficients of determination because the number of degrees of freedom is 4, while the number of observations in the dataset is 116, leading to obtaining

$$\frac{116-1}{116-4} = 1.02678$$

On the contrary, if we consider a linear regression model that includes the following explanatory variables

- **Weight**
- **Horsepower**
- **Displacement**
- **Turning Circle**

together with their squares and their pairwise interactions, we will obtain the following values of the coefficients of determination:

$$R^2 = 0.80808$$

$$R^2_{adjusted} = 0.78148$$

In this case, we begin to observe a discrepancy between the two coefficients; the increase in model complexity (degrees of freedom) might not be justified by the increase in the value of the coefficient of determination.

Note that in simple linear regression models, it is possible to verify that the coefficient of determination coincides with the square of the linear correlation coefficient.

In the case where the simple linear regression model has the variable Gas Tank Size as the target variable and the variable Weight as the explanatory variable, we obtain the following:

$$R^2 = 0.71707$$

$$corr(\text{Gas Tank Size}, \text{Weight}) = 0.8468$$

MULTI-COLLINEARITY

Multi-collinearity of independent variables

In a multiple linear regression model, it is required that the explanatory variables are independent of each other.

If a significant linear correlation is found between two or more independent variables, then it is referred to as collinearity or multicollinearity.

Under such conditions, the estimation of the coefficients becomes inaccurate and compromises the significance of the regression model.

It is important to address the causes of collinearity, for example, by selecting only certain explanatory variables or only certain terms in which they appear. One way to identify if there is multicollinearity among the explanatory variables is to compute the correlation matrix between them and conclude that two of them are collinear if the absolute value of the correlation is greater than a certain threshold.

In the case of the quadratic regression model for predicting the target variable 'Gas Tank Size,' we will obtain the following correlation matrix

	Weight	Horsepower	Displacement	Turning Circle
Weight	1.0000	0.7067	0.8300	0.7518
Horsepower	0.7067	1.0000	0.7642	0.4825
Displacement	0.8300	0.7642	1.0000	0.7467
Turning Circle	0.7518	0.4825	0.7467	1.0000

To highlight multivariate relationships among the explanatory variables, it is possible to compute the Variance Inflation Factor (VIF) defined for each variable X^j as follows

$$VIF_j = \frac{1}{1 - R_j^2}$$

where R^2_j represents the coefficient of determination of the linear regression model that has the variable X^j as the response variable and the remaining explanatory variables of the original regression model as explanatory variables.

Typically, VIF values greater than 5 indicate the presence of multicollinearity among the explanatory variables.

CONFIDENCE LIMITS AND PREDICITONS

Once we have validated the regression model

$$Y = f\left(X^1,\dots,X^n\right)$$

It can be use to predict the expected value of the target variable Y for a specific instance of the vector of explanatory variables $\underline{X}$

$$\underline{x} = \left(x^1,\dots,x^n\right)$$

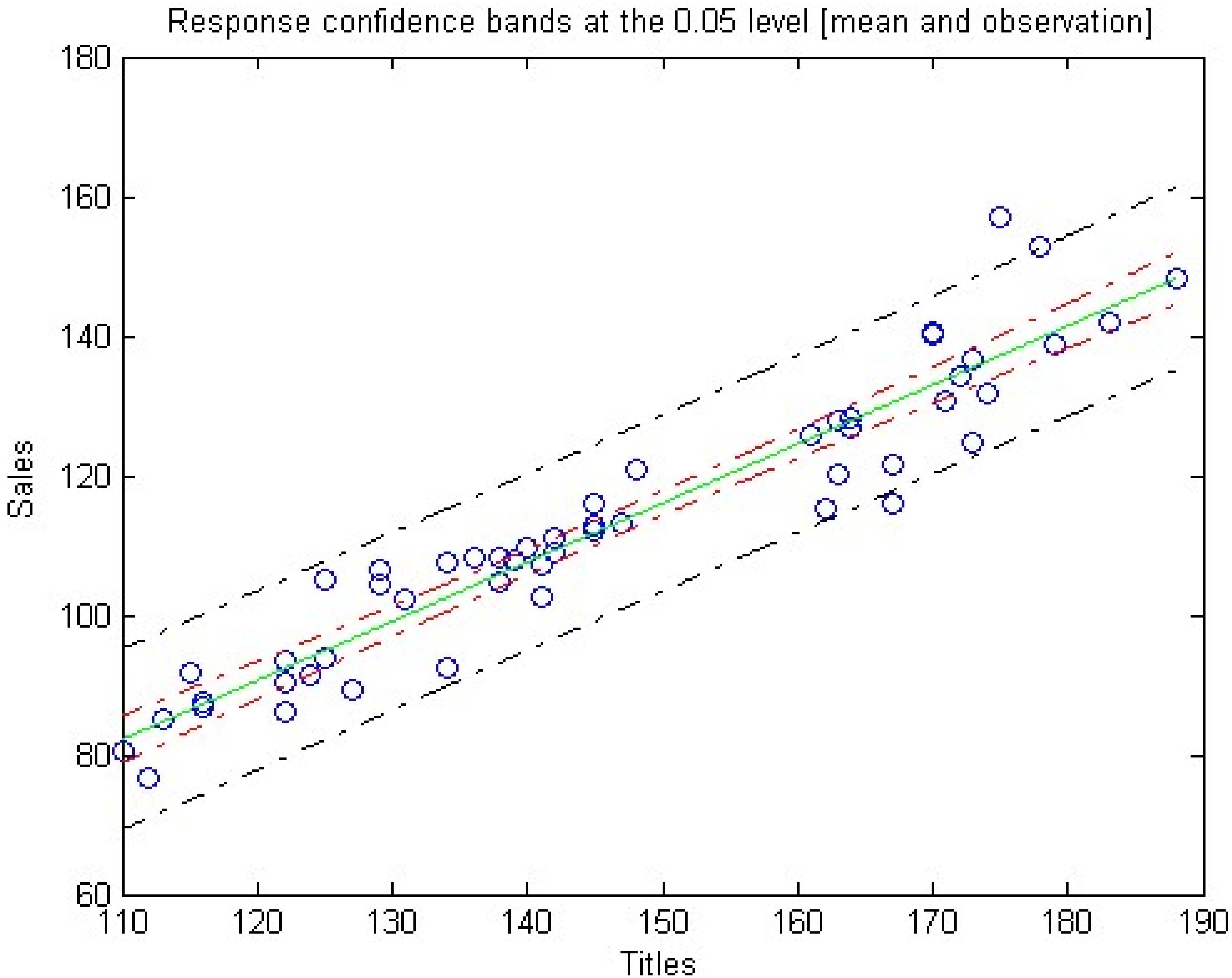
The regression model gives nin only the expected value of the response variable Y but also some insights on the utility of the model itself:

- *prediction interval* — *contains the expected value* ***E[Y]***
- *confidence interval* — *contains the value* ***Y***

For any given instance $\underline{x}$ of the vector of explanatory variables $\underline{X}$.

In the case of the linear regression model for the target variable 'Sales' that uses the variable 'Titles' as the explanatory variable, we obtain the graph shown in the following slide.

CONFIDENCE LIMITS AND PREDICITONS



VARIABLE SELECTION

The development of a multiple linear regression model requires selecting a subset of all candidate variables that are the most informative for explaining the behavior of the target variable.

The selection of predictive variables is a particularly complex task, especially when the number of candidate variables to be used as explanatory variables is very high.

To this end, automatic variable selection techniques have been developed, which are divided into

- *Forward inclusion*
- *Backward elimination*
- *Mixed*

Forward Inclusion

Starting from a solution where the set of selected variables is empty, a variable is added to the set of selected variables by taking it from the set of candidate variables to be explanatory variables (without repetition).

Among all the variables not yet added to the set of selected variables, the one that provides the maximum increase in predictive value for the target variable is included.

There are different criteria for choosing the variable to include, among which we can mention:

- *reduction of the sum of squared residuals*
- *increase in the adjusted regression coefficient*
- *increase in the value of the F statistic*

VARIABLE SELECTION

Backward Exclusion

Starting from a solution where the set of selected variables contains all the candidate variables to be explanatory variables, a variable is removed from the set of selected variables.

Among all the variables not yet removed from the set of selected variables, the one that results in the smallest reduction in predictive value for the target variable is eliminated.

There are different criteria for choosing the variable to exclude.

VARIABLE SELECTION

If we apply the Forward Inclusion procedure to the linear regression model with the target variable being Gas Tank Size and the following variables as candidate explanatory variables: Weight, Horsepower, Turning Circle, and Displacement with the following parameters:

- *Prob to enter = 0.025*
- *Maxiterations = 100*

We get

```
Initial columns included:  none
Step 1, added column 1, p=4.89546e-033
Final columns included:  1
    'Coeff'      'Std.Err.'                    'Status'        'P'
    [ 0.0049]    [2.8604e-004]                 'In'            [4.8955e-033]
    [ 0.0104]    [     0.0054]                 'Out'           [     0.0558]
    [-0.0418]    [     0.0744]                 'Out'           [     0.5758]
    [ 0.0026]    [     0.0046]                 'Out'           [     0.5650]
```

VARIABLE SELECTION

If we apply the Backward Elimination procedure to the same linear regression model with the same parameter values, we obtain

```
Initial columns included:  1 2 3 4
Step 1, removed column 4, p=0.860955
Step 2, removed column 3, p=0.713442
Step 3, removed column 2, p=0.0558067
Final columns included:  1
    'Coeff'          'Std.Err.'        'Status'          'P'
    [ 0.0049]        [2.8604e-004]     'In'         [4.8955e-033]
    [ 0.0104]        [     0.0054]     'Out'        [     0.0558]
    [-0.0418]        [     0.0744]     'Out'        [     0.5758]
    [ 0.0026]        [     0.0046]     'Out'        [     0.5650]
```

Interestingly, both Stepwise Selection procedures lead to including only the first variable in the regression model, assessing that only Weight is informative for predicting the value of the target variable Gas Tank Size